# REPORT

Reviewer report: pytoolz/toolz#666: Keep

EqualityHashKey comparisons consistent with hashing

Run toolz-666, tested 2026-11-14 12:47 IST by ops/daily_run.py (route source-pair).
Proof-of-Change commit

ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-14 12:58 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

COULD NOT VERIFY

Recommended action: Re-run Confidence: Low

The run did not produce a usable comparison, so there is no conclusion about the PR. The run did not finish

cleanly, so it says nothing about the PR.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: Keep EqualityHashKey comparisons consistent with hashing (pytoolz/toolz#666).

• What we observed: not verified: no AI-written probe ran any changed line, so the change itself was not

exercised by a probe; 2 test(s) fail on main and pass on the PR; 6 test(s) ran on main, 8 on the PR; 2 new test(s)

pass, 4 unchanged.

• What you should do: Do not decide yet. Fix the problem under 'How to fix' and run the check again.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 10.4 GB (of 15.6 GB).

• Setup check: free disk 92.4 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: toolz python setup install instructions https://github.com/pytoolz/toolz site:github.com/pytoolz/toolz

• Query: toolz python documentation install https://github.com/pytoolz/toolz site:github.com/pytoolz/toolz

• Query: toolz python install requirements https://pypi.org/project/toolz/ https://github.com/pytoolz/toolz

site:pypi.org

• Source: toolz/README.rst at master · pytoolz/toolz · GitHub —

https://github.com/pytoolz/toolz/blob/master/README.rst

• Source: toolz/README.rst at master · pytoolz/toolz · GitHub —

http://github.com/pytoolz/toolz/blob/master/README.rst

• Source: Toolz — https://github.com/pytoolz/toolz/raw/refs/heads/master/README.rst

• Source: toolz/README.rst at 451af60dec590a6010e2babdbf391ea8f815122f · pytoolz/toolz · GitHub —

https://github.com/pytoolz/toolz/blob/451af60dec590a6010e2babdbf391ea8f815122f/README.rst

• Source: toolz — https://pypi.org/project/toolz/

• Setup plan: the same packages on main and pr.

• Setup plan: install coverage (listed in the test extra) on main and pr.

• Setup plan: install pycodestyle (listed in the test extra) on main and pr.

• Setup plan: install pytest (listed in the test extra) on main and pr.

• Setup plan: install sphinx (listed in requirements.txt) on main and pr.

• Setup plan: install furo (listed in requirements.txt) on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.9

• Installed coverage 7.17.3 from pyproject (listed in the test extra) on main and pr.

• Installed pycodestyle 2.15.0 from pyproject (listed in the test extra) on main and pr.

• Installed pytest 9.2.0 from pyproject (listed in the test extra) on main and pr.

• Installed sphinx 9.1.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed furo 2025.12.19 from requirements.txt (listed in requirements.txt) on main and pr.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| C1 | Keep EqualityHashKey comparisons consistent with hashing (PR title) | not testable | claims are read out of

the PR description only by model-written probes (--model-provider, see Probes); the tests below check behavior |

pytest-pr-1.log:7 «8 passed in 0.75s»

| C2 | Existing behavior is kept: tests that pass on main still pass (implied by every PR) | proven | no test that

passes on main fails on the PR (4 unchanged, 3 runs per side) | pytest-pr-1.log:7 «8 passed in 0.75s»

Citation check (PASS): 2 citation(s), every one re-read from its log file and matched.

What changed in behavior

| What | Count | Examples

| Tests that actually ran (main / PR, fewest of the runs) | 6 / 8 | | Tests unchanged | 4 | | New tests that pass | 2 | | Tests that pass on main but fail on the PR | 0 | | Tests that are new in the PR and fail | 0 | | Tests that ran on main but are missing on the PR side | 0 | | Tests that changed result in another way (for example passed -> skipped) | 0 | | Tests that gave different results between the 3 runs of the same side (flaky) | 0 | | Tests that fail on main and pass on the PR | 2 |

toolz.sandbox.tests.test_equality_hash_contract::test_different_hash_strategies_do_not_compare_equal,

toolz.sandbox.tests.test_equality_hash_contract::test_unrelated_operands_can_use_reflected_equality

| Tests that are new or changed but did not run because an optional module is missing | 0 | | Tests that were re-parametrized (same test, new parameter ids; not counted as removed) | 0 | | Test modules that fail to load on the PR side only (their tests no longer run) | 0 | | Test modules that fail to load on main only (no baseline for their tests) | 0 | | Test modules that fail to load on both sides (their tests did not run) | 0 | | Tests that fail identically on both sides (already broken on main) | 0 | -

Probes

Model-written probes: model provider: nebius (Token Factory, live).

No probe ran for this PR (no fixed probe in probes/<owner>/<repo>/<pr>.py and no model-written or differential

probe), so only the tests were compared.

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (1.6 s run).

Coverage gate (python, confidence normal): 5 of 5 changed executable line(s) ran (100%).

| Changed file | Changed executable lines | Of those, ran

| toolz/sandbox/core.py | 5 | 5

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested b500f2b26da0 of the PR against main (source-pair route).

• 3 alternating runs per side: main 3, PR 3 of 3.

• pytest exit codes explained: all runs exited 0.

• Changed code was exercised (coverage gate): 5 of 5 changed executable line(s) ran (100%).

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every claim cites a real log line: 2 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

| Issue | Status | Detail | How to fix

| Run did not finish | COULD NOT VERIFY | not verified: no AI-written probe ran any changed line, so the change

itself was not exercised by a probe; 2 test(s) fail on main and pass on the PR; 6 test(s) ran on main, 8 on the PR;

2 new test(s) pass, 4 unchanged | Read the setup/build log under Evidence links; build failures usually need a

compiler or a newer nightly.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | PASS | main 3, PR 3 of 3

| pytest exit codes explained | PASS | all runs exited 0

| Changed code was exercised (coverage gate) | PASS | 5 of 5 changed executable line(s) ran (100%)

| Probes ran on both sides | SKIP | no probe for this PR

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

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| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

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Checklist

• I read the verdict and the plain-English summary.

• Every 'false' or 'unmentioned change' row has a log line I can open.

• Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).

• I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide REPORT=poc-main/reports/daily/2026-11-14/toolz-666 DECISION=approve (or reject, approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/pytoolz/toolz/pull/666

Evidence links

• Pull request pytoolz/toolz#666

• record.json

• pr.diff

• setup-main.log

• setup-pr.log

• git.log

• pytest-main-1.log

• pytest-pr-1.log

• pytest-main-2.log

• pytest-pr-2.log

• pytest-main-3.log

• pytest-pr-3.log

• junit-main-1.xml

• junit-pr-1.xml

• coverage-pytest.log

• coverage-pr.json

• main run 1 result: pytest-main-1.log:7 «4 passed in 0.68s»

• pr run 1 result: pytest-pr-1.log:7 «8 passed in 0.75s»

Glossary

| Term | Meaning

| main | The library as it is today, without the PR.

| PR | Pull request: a proposed change to the library.

| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.

| flaky | A test that gives different results when nothing changed; it is ignored as evidence.

| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-14 12:59 IST